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A COVID-19 PATIENT WITH COMBINED BILATERAL ADRENAL HEMORRHAGE

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A 75-year-old female patient with fever and cough showed positive for nucleic acid antibodies. First chest CT (Fig.1 A1-A3) showed novel coronavirus pneumonia (COVID-19) and bilateral adrenal glands were normal. Due to left abdomen pain after 2 weeks, chest CT and abdominal CT enhanced scan showed expanded infection in bilateral lungs and bilateral adrenal hemorrhages (AH) (Fig.1 B1-B3, Fig. 2). Laboratory tests: plasma D-dimers elevated (3.11 mg/L), monocyte count elevated (4.27×10^9/L), norepinephrine increased 6.38 μmol/24 hours, serum ACTH increased (73pg/ml), serum cortisol and serum sodium decreased (98nmol/L, 123mmol/L, respectively). Blood pressure was 100/60mmHg. After 2 weeks chest and abdominal CT (Fig. 1 C1-C3) indicated infection reduce in bilateral lungs. Adrenal hemorrhage reduced in right side and increased in left side. COVID-19 was treated with hormones (methylprednisolone, qd 40 mg, 5 days).

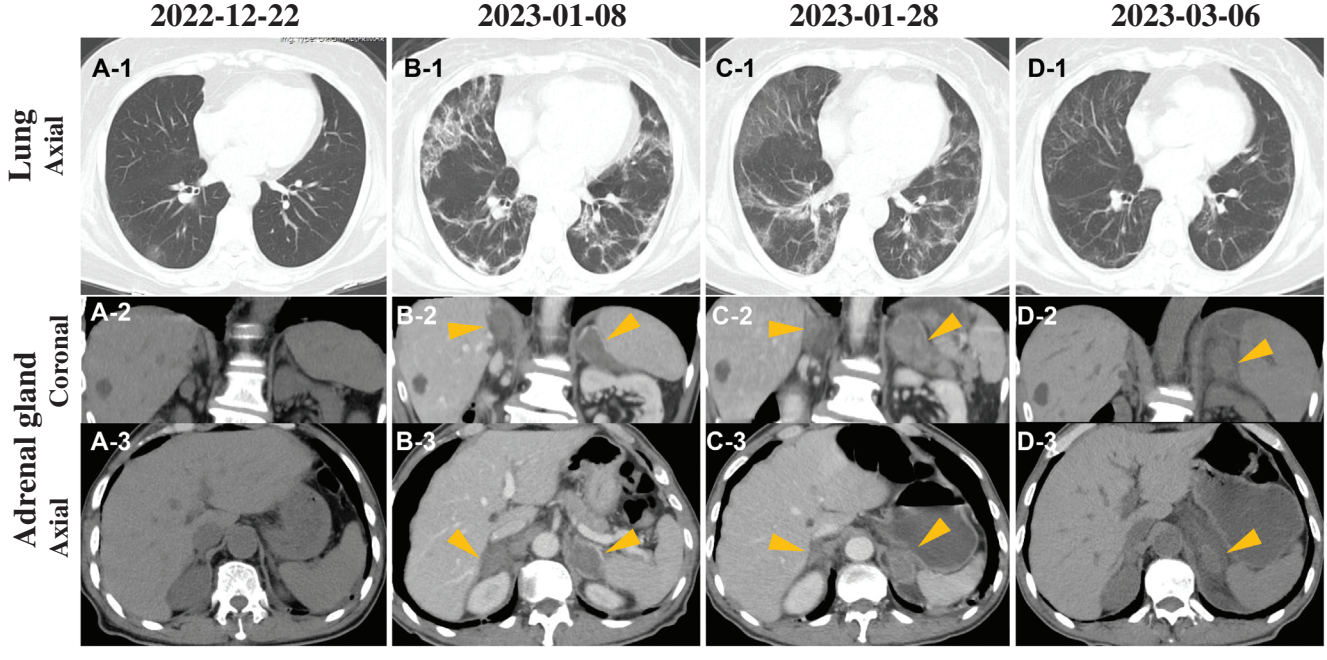


Figure 1. CT manifestations of the course of the disease in this patient. 2022-12-22: Little infection in both lungs (A-1), normal bilateral adrenal glands (coronal: A-2, axial: A-3). 2023-01-08: Infection enlargement in both lungs (B-1), bilateral adrenal hemorrhage (AH) (coronal: B-2, axial: B-3; arrows), right AH: 3.0 cm × 1.8 cm × 2.8 cm; left AH: 5.4 cm × 2.4 cm × 3.3 cm. 2023-01-28: Infection reduction in both lungs (C-1), changes in bilateral AH (coronal: C-2, axial: C-3; arrows), right AH reduction (2.4 cm × 1.2 cm × 1.6 cm), left AH enlargement (5.5 cm × 3.0 cm × 3.6 cm). 2023-03-06: Bilateral pulmonary restrictive fibrosis (D-1), bilateral AH shrinkage (coronal: D-2, axial: D-3; arrows) (right: 1.6 cm × 1.2 cm × 1.0 cm, left: 2.3 cm × 1.5 cm × 2.1 cm).

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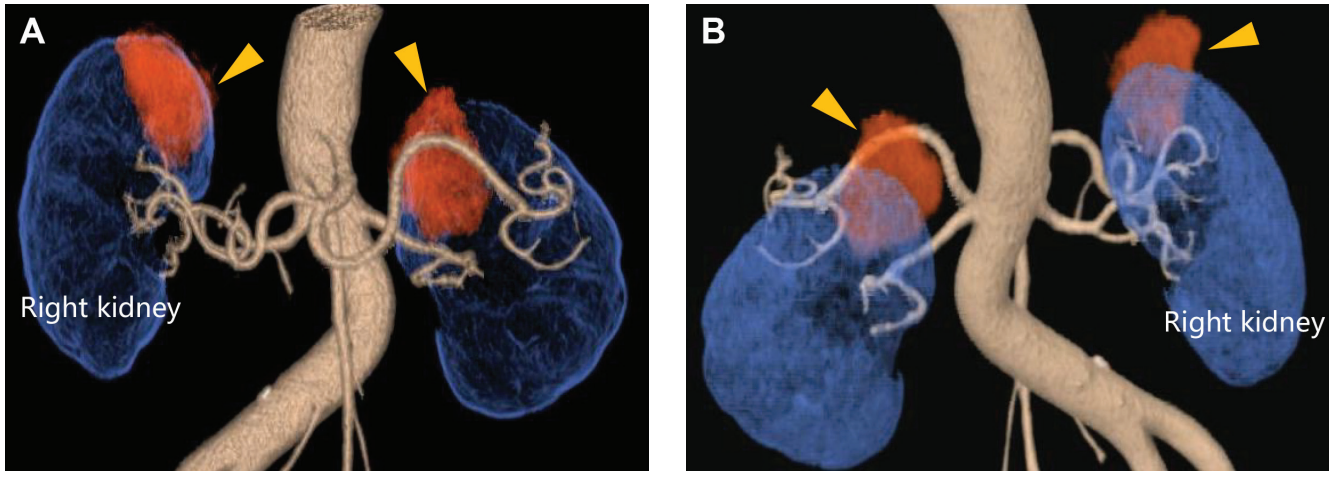


Figure 2. CT Volume rendering (VR) technique on bilateral adrenal hemorrhages (arrows), anterior view (A) and posterior view (B).

Hormone replacement was used to treat the reduction of cortisol (Ponisone, bid am. 5mg and pm. 2.5mg, 1 month). After hormonal, anticoagulant, hemostatic and anti-viral treatment, the patient was discharged from the hospital. Re-examination showed bilateral adrenal hemorrhages reduced after 2 months (Fig.1 D1-D3).

AH caused by COVID-19 is a rare extrapulmonary complication (1) and can lead to acute adrenal insufficiency and even death. Currently, the pathogenesis of AH may be: (i) vascular rupture bleeding due to the increased adrenal vein pressure (2), (ii) vascular rupture caused by adrenal vascular endothelial cell injury (3), (iii) immune response that destroys adrenal tissue (4). Neurological manifestations of Covid-19 do also innitiate this workup (5).

It was always neglected because the atypical symptoms of AH in clinic.

Conflict of interest

The authors declare that they have no conflict of interest.

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